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Semester IV (B.Tech)

Er. No..................

Academic Year: 2022-23

Jaypee University of Engineering & Technology, Guna

T-1(Even Semester 2023)

18B11CI415/14B11CI711 – ARTIFICIAL INTELLIGENCE & APPLICATIONS

Maximum Duration: 1 Hour

Maximum Marks: 15

Notes:

1. This question paper has four questions.
2. Write relevant answers only.
3. Do not write anything on question paper (Except your Er. No.).
4. **Answer the questions in serial order.**

		Marks	CO No.
Q1.	On the Basis of 'Human-Like approach' and 'Rationality' compare and contrast between Cognitive science Approach & Laws of thought Approach also Turing test Approach & Rational agent Approach.	[03]	CO2
Q2.	Draw the AND-OR tree/graph structure to represent the following facts: Improve enjoyment of life, Improve standard of living, Work less hard, Provide for old age, Save money, Earn more money, Go on strike, Improve productivity.	[03]	CO2
Q3.	If all the hard constraints are linear and some are inequalities, but the objective function is quadratic, the problem is a quadratic programming problem. It is one type of nonlinear programming. By which method this problem can be solved? You have to typically name the method and justify it.	[03]	CO3
Q4.	For the belief system shown below in (Fig1), and the corresponding conditional probability tables (Fig2) compute: P(AE'), P(A'E'), P(JB), P(J | B), P(MB), P(M | B).	[06]	CO3

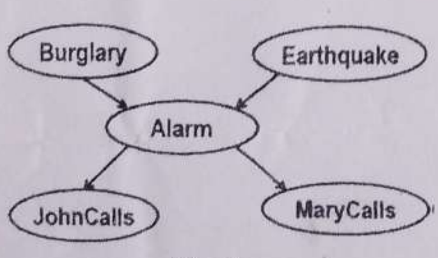

Fig 1.

B	E	P(A)
T	T	0.95
T	F	0.95
F	T	0.29
F	F	0.001

A	P(J)
T	0.90
F	0.05

A	P(M)
T	0.70
F	0.01

P(E)
0.002

P(B)
0.001

0.998 0.999

Fig 2.

Semester IV (B.Tech)

Er. No...................
Academic Year: 2022-2

Jaypee University of Engineering & Technology, Guna

T-1(Even Semester 2023)

18B11CI412 – ALORITHMS AND PROBLEM SOLVING

Maximum Duration: 1 Hour
15

Maximum Marks:

Notes:

1. This question paper has ***three*** questions.
2. Write relevant answers only.
3. Do not write anything on question paper (Except your Er. No.).

Marks

Q1. (a) Write a recursive function for the running time T(n) of the function given below. Prove using iterative method that $T(n) = O(n^3)$. **[03]**

```
def function(n):
    count=0
    if n<=0:
        return
    for i in range (0, n):
        for j in range (0, n):
            count=count+1
    function(n-3)
    print(count)
```

(b) Suppose we are sorting an array of eight integers using heap-sort, and we have just finished some heapify (either max-heapify or min-heapify) operations. The array now looks like this: 16 14 15 10 12 27 28. How many heapify operations have been performed on root of heap? Justify your outcome. **[02]**

Q2. (a) A 3-ary max heap is like a binary max heap, but instead of 2 children, nodes have 3 children. A 3-ary heap can be represented by an array as follows: The root is stored in the first location, a[0], nodes in the next level, from left to right, is stored from a[1] to a[3]. The nodes from the second level of the tree from left to right are stored from a[4] location onward. An item x can be inserted into a 3-ary heap containing n items by placing x at the location a[n] and pushing it up the tree to satisfy the heap property. Design and discuss efficient procedure to **build 3-ary max heap.** **[03]**

(b) Can the master method be applied to the recurrence $T(N) = 4T(N/2) + N^2 \log N$? Why or why not? Give an asymptotic upper bound for this recurrence. [02] CO4

Q3. (a) Illustrate the operation of COUNTING-SORT on the array A =(6,0,2,0,1,3,4, 6,1,2). Suppose that we rewrite the for loop header in line 10 of the COUNTING SORT as **10. for j = 1 to A.length** [03] CO2

Show that the algorithm still works properly. Is the modified algorithm stable?

(b) illustrate the operation of RADIX-SORT on the following list of English words: COW, DOG, SEA, RUG, ROW, MOB, BOX, TAB, BAR, EAR, TAR [02] CO2

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Semester IV (B.Tech)

Er. No.....................

Academic Year: 2022-23

Jaypee University of Engineering & Technology, Guna

T-1 (Even Semester 2023)

18B14HS441- Concept of Digital Marketing

Maximum duration: 1 Hour

Maximum Marks: 15

Notes:

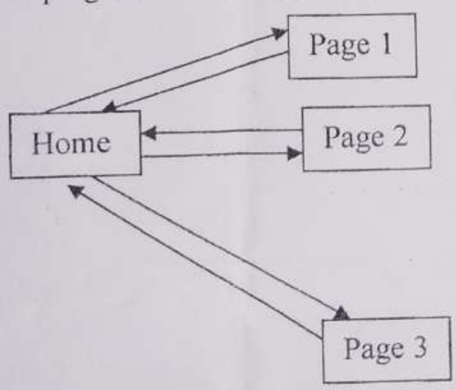
1. This question paper has 3 questions.
2. Write relevant answers only.
3. Do not write anything on question paper except Enroll No.

Q1. 'We all have witnessed the inseparable revolution of technology as well as the importance of Digital Marketing. Increased accessibility and affordability of electronic media have given immense power to this emerging technology of Digital Marketing". Would you argue that this statement is largely true or false? Why or why not? Differentiate between digital marketing over traditional marketing to justify your argument. **[05]** CO2

Q2. (a) 'According to the research consultancy IDC, the technology industry is expected to reach $5 trillion in 2021 which represents 4.2% growth, signalling a return to the trend line that the industry was on prior to the pandemic.' With flexibility and resilience as the guiding principles for future success, organizations will adopt a cloud-first mentality when it comes to building or upgrading IT infrastructure. Businesses are placing much more emphasis on strategic IT as opposed to the tactical mindset of previous decades. This means that technology is a driver for business objectives rather than simply playing a supporting role. In view of the facts related to IT industry, analyze the IT industry for entrepreneurs to shape their strategy to drive profitability using Porter's five forces model. **[03]** CO4

(b) 'Digital marketing is a valuable asset to your business's growth and helps you establish an authoritative online presence'. In lieu of the statement, explicate the process of digital marketing. **[02]** CO2

Q3. Using the graph shown below, compute the PageRank at every node at the end of third iteration. Use damping factor=0.85. (Take upto two decimals) **[05]** CO3



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Semester IV (B.Tech) **Er. No..................**

Academic Year: 2022-23

Jaypee University of Engineering & Technology, Guna

T-1 (Even Semester 2023)

18B11CI411 – COMPUTER NETWORKS

Maximum Duration: 1 Hour Maximum Marks: 15

Notes:

1. This question paper has three questions.
2. Write relevant answers only.
3. Do not write anything on question paper (Except your Er. No.).

		Marks	CO No.
Q1.	Write the functions and protocols of all the layers of TCP/IP Model.	[03]	CO2
Q2.	Draw the waveforms of different techniques for converting digital to analog signal used for long distance transmission. A signal has passed through three cascaded amplifiers, each with a 6 dB gain and loss of 2dB in links between them. What is the total gain/loss? How much is the signal amplified.	[03]	CO3
Q3.	Discuss different line coding schemes. Draw the waveforms for bit stream 110100 using Unipolar NRZ, Polar RZ and Manchester schemes.	[03]	CO2
Q4.	What are the various components of latency in telecommunication networks? Calculate the total latency for a frame of size 2 million Bytes that is being sent on a link with 10 routers each having a queuing time of 2 μsec and a processing time of 1 μsec. The length of link is 2000 km and the speed of light inside link is 2×10^8 m/s. The link has bandwidth of 4 Mbps. Which component of total delay is dominant?	[03]	CO3
Q5.	Illustrate the transmissions impairments and how they are calculated? Write down the addressing modes used in telecommunication networks.	[03]	CO4

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Semester IV (B. Tech)

Er. No. [redacted]........
Academic Year: 2022-23

Jaypee University of Engineering & Technology, Guna

T-1(Even Semester 2023)

18B11CI413 – OPERATING SYSTEMS

Maximum Duration: 1 Hour

Maximum Marks: 15

Notes:

1. This question paper has THREE questions.
2. Write relevant answers only.
3. Do not write anything on question paper (Except your Er. No.).

		Marks	CO No.
Q1.	Draw the process state transition diagram and explain the following situation: Consider two processes running, each of which uses the CPU. The first process issues an I/O after running for some time.	[05]	CO2
Q2.	The following processes are being scheduled using a preemptive-priority-based-round-robin scheduling algorithm.		CO3

Process	Burst Time	Arrival Time	Priority
P_1	8	0	0
P_2	3	5	5
P_3	4	6	20
P_4	4	7	20
P_5	5	9	22

Each process is assigned a numerical priority, with a lower number indicating a higher relative priority. The scheduler will execute the highest priority process. For processes with the same priority, a round-robin scheduler will be used with a time quantum of 2 units, otherwise default time slice is 5ms. If a process is preempted by a higher-priority process, the preempted process is placed at the first of the queue.

(a) Show the scheduling order of the processes using a Gantt chart. [02]

(b) What is the turnaround time for each process? [01]

(c) What is the average waiting time.? [01]

(d) What is the response time for each process? [01]

Q3. The following processes are being scheduled using a Completely Fair Scheduler algorithm. CO3

Process	Burst Time	Priority	Weight
P_1	13	0	1024
P_2	18	-5	3121
P_3	22	-4	2501
P_4	23	-2	1586
P_5	25	-3	1991

Each process priority and weight value are given in above table.

(a) Calculate the time slice of each process. [01]

(b) Show the scheduling order of the processes using a Gantt chart. [02]

(c) What is the turnaround time for each process? [01]

(d) What is the average waiting time.? [01]

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